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202606UECM3473OE2a

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Review of preview

Started on	Monday, 13 July 2026, 04:02 PM
Completed on	Monday, 13 July 2026, 04:03 PM
Time taken	54 secs
Marks	0/9
Grade	0 out of a maximum of 10 (0%)

1 

Marks: 1

You are given:

- The number of claims for each individual follows Poisson distribution with mean λ .
- λ varies by individual in accordance with gamma distribution with parameters $\alpha = 2$ and $\theta = 1.57$.
- Claims severity follow a distribution with probability density function: $f(x) = 1.4^7/\Gamma(7)x^{-8}e^{-1.4/x}$, $x > 0$.
- The standard for full credibility of aggregate loss experience is set so that the probability of observed claims being within 9% of expected claims is 99%.

Answer:



[Make comment or override grade](#)

Incorrect

Correct answer: 2269.27

Marks for this submission: 0/1.

2



Marks: 1

You are given:

- The number of claims has a Poisson distribution with mean 0.02.
- Claims sizes have a log normal distribution distribution with $\sigma = 1.2$.
- The number of claims and claim sizes are independent.
- The observed pure premium should be within 2% of the expected pure premium 99% of the time.

Determine the expected number of exposures needed for full credibility. _____

Answer:



[Make comment or override grade](#)

Incorrect

Correct answer: 3500949

Marks for this submission: 0/1.

3



Marks: 1

You are given:

- The number of claims follows a negative binomial distribution with parameters r and $\beta = 5$.
- Claim severity has the following distribution:

Claim Size	Probability
1	0.32
10	0.35
100	0.33

- The number of claims is independent of the severity of claims.

Determine the expected number of claims needed for aggregate losses to be within 9% of expected aggregate losses with 95% probability.

Answer:

X

[Make comment or override grade](#)

Incorrect

Correct answer: 3538.16

Marks for this submission: 0/1.

4



Marks: 1

For a group dental plan, each individual's number of claims follow Poisson distribution with parameter λ . λ varies by individual in accordance with the following distribution:

λ	Probability
1	0.57
2	0.23
5	0.20

Claim sizes follow log normal distribution with parameter μ and $\sigma = 0.30$.

Classical credibility techniques are used.

The standard for full credibility of aggregate loss experience is set so that the probability of observed claims being within 5.90% of expected claims is 99%. Determine the number of claims required for full credibility. _____

Answer:

X

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Incorrect

Correct answer: 4310.53

Marks for this submission: 0/1.

5



Marks: 1

The full credibility standard for a company is set according to the methods of classical credibility so that the total number of claims is to be within 1% of the true value with probability P . This full credibility standard is calculated to 817 claims. The standard is altered so that the total cost of claims is to be within 2% of the true value with probability P . The claim frequency has a Poisson distribution and the claim severity had the distribution

$$f(x) = (120-x)/7200.0, \text{ for } 0 < x < 120$$

What is the expected number of claims necessary to obtain full credibility under the new standard? _____

Answer:



[Make comment or override grade](#)

Incorrect

Correct answer: 306.375

Marks for this submission: 0/1.

6



Marks: 1

Aggregate claims follows a Pareto distribution with parameters $\alpha = 5$ and $\theta = 8$. The full credibility standard is set according to the methods of classical credibility so that actual aggregate claims are within 10% of expected aggregate claims 95% of the time. Determine the amount of expected aggregate claims needed for full credibility. _____

Answer:



[Make comment or override grade](#)

Incorrect

Correct answer: 1280.533333

Marks for this submission: 0/1.

7



Marks: 1

You are given the following:

- 125,100 exposures are needed for full credibility.
- The 125,100 exposures standard was selected using a normal approximation so that the actual total cost of claims is within 7.5%; of the expected total 95%; of the time.
- The number of claims per exposure follows a Poisson distribution with mean m .
- m was estimated from the following observed data using the maximum likelihood:

Year	Exposures	Number of Claims
1	17,714	1,177
2	25,671	1,710
3	22,904	1,375

If mean claim severity is 1,460, determine the standard deviation of the claim severity distribution. _____

Answer:



[Make comment or override grade](#)

Incorrect

Correct answer: 4792.967976

Marks for this submission: 0/1.

8 

Marks: 1

For an insurance coverage you are given:

- Claim counts follow a Poisson distribution.
- Claim sizes follow an exponential distribution with mean μ .
- μ varies by insured according to a gamma distribution with parameters $\alpha = 8$ and $\theta = 230$.

The methods of limited fluctuation credibility are used. 2900 expected claims are required for full credibility. The full credibility standard is that actual claims should be within 1% of expected claims with probability p . Determine p . _____

Answer:

X

[Make comment or override grade](#)

Incorrect

Correct answer: 0.2812

Marks for this submission: 0/1.

9 

Marks: 1

You are given:

- Claim frequency has a Poisson distribution.
- Claims size has a Gamma distribution with $\alpha = 2.5$, θ unknown.
- Using the methods of classical credibility, a full credibility standard of 1170 expected claims has been established so that actual aggregate claim costs will be within 8% of expected aggregate claim costs $P\%$ of the time.

Determine P . _____

Answer:

X

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Incorrect

Correct answer: 97.92

Marks for this submission: 0/1.

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